

Java Naming Conventions

- **Java naming convention is a rule to follow as you decide what to name your identifiers such as class, package, variable, constant, method, etc.**
- But, it is not forced to follow. So, it is known as convention not rule.
- These conventions are suggested by several Java communities such as Sun Microsystems and Netscape.

All the classes, interfaces, packages, methods and fields of Java programming language are given according to the Java naming convention. If you fail to follow these conventions, it may generate confusion or erroneous code.

Java uses CamelCase as a practice for writing names of methods, variables, classes, packages and constants.

Camel case in Java Programming : It consists of compound words or phrases such that each word or abbreviation begins with a capital letter or first word with a lowercase letter, rest all with capital.

1. **Classes and Interfaces :**

- Class names should be **nouns**, in mixed case with the **first** letter of each internal word capitalised. Interfaces name should also be capitalised just like class names.
- Use whole words and must avoid acronyms and abbreviations.

Examples:

```
interface Bicycle
class MountainBike implements Bicycle

interface Sport
class Football implements Sport
```

2. **Methods :**

- Methods should be **verbs**, in mixed case with the **first letter lowercase** and with the first letter of each internal word capitalised.

Examples:

```
void changeGear(int newValue);
void speedUp(int increment);
void applyBrakes(int decrement);
```

3. **Variables** : Variable names should be short yet meaningful.
- Should **not** start with underscore('_') or dollar sign '\$' characters.
 - Should be mnemonic i.e, designed to indicate to the casual observer the intent of its use.
 - **One-character variable names should be avoided** except for temporary variables.
 - Common names for temporary variables are i, j, k, m, and n for integers; c, d, and e for characters.

Examples:

```
// variables for MountainBike class  
int speed = 0;  
int gear = 1;
```

4. **Constant variables:**

- Should be **all uppercase** with words separated by underscores ("_").
- There are various constants used in predefined classes like Float, Long, String etc.

Examples:

```
static final int MIN_WIDTH = 4;  
  
// Some Constant variables used in predefined Float class  
public static final float POSITIVE_INFINITY = 1.0f / 0.0f;  
public static final float NEGATIVE_INFINITY = -1.0f / 0.0f;  
public static final float NaN = 0.0f / 0.0f;
```

5. **Packages:**

- The prefix of a unique package name is always written in **all-lowercase ASCII letters** and should be one of the top-level domain names, like com, edu, gov, mil, net, org.
- Subsequent components of the package name vary according to an organisation's own internal naming conventions.

Examples:

```
com.sun.eng  
com.apple.quicktime.v2  
  
// java.lang packet in JDK  
java.lang
```